NAME - Montu Jangid

Assignment – Malware and Threat Detection

Module 3 – CS: Cyber Threats & CEH

---

1. What are the Different Types of Hacking Methods?

Answer -

Hacking methods are techniques used by attackers to gain unauthorized access to systems, networks, or data. Different methods are used depending on the target and purpose of the attack.

Common Types of Hacking Methods

1. Phishing

Phishing is a method where fake emails, messages, or websites are used to trick users into sharing passwords, banking details, or personal information.

2. Password Attacks

Attackers try to guess or crack passwords using different techniques such as brute force or dictionary attacks.

3. Malware Attacks

Malicious software such as viruses, worms, ransomware, and spyware are used to damage systems or steal data.

4. Social Engineering

Attackers manipulate people psychologically to gain confidential information or system access.

5. Denial of Service (DoS) Attack

A DoS attack floods a system or website with excessive traffic, making it unavailable to users.

6. Man-in-the-Middle (MITM) Attack

In this attack, the attacker secretly intercepts communication between two users to steal information.

---

2. Explain Types of Password Attacks

Answer -

Password attacks are methods used by attackers to obtain user passwords and gain unauthorized access to systems.

Types of Password Attacks

1. Brute Force Attack

The attacker tries every possible password combination until the correct password is found.

2. Dictionary Attack

A list of common passwords and words is used to guess the password.

3. Credential Stuffing

Previously leaked usernames and passwords are used to access multiple accounts.

4. Keylogging

A keylogger records keyboard activity to capture passwords entered by users.

5. Phishing Attack

Fake login pages or emails are used to trick users into revealing passwords.

Prevention Methods

- Use strong passwords
- Enable multi-factor authentication
- Avoid sharing passwords
- Use password managers

---

3. Explain Password Cracking Tools: pwdump7, Medusa and Hydra

Answer -

1. pwdump7

pwdump7 is a password recovery and auditing tool used to extract password hashes from Windows systems. Security professionals use it to test password strength.

2. Medusa

Medusa is a network login brute-force testing tool that supports many protocols such as SSH, FTP, and HTTP. It is used for password auditing and security testing.

3. Hydra

Hydra is a fast password-cracking tool used to test login credentials on different network services. It supports multiple protocols and is commonly used in penetration testing.

Note

These tools should only be used for ethical security testing and authorized purposes. Unauthorized use is illegal.

---

4. Explain Types of Steganography with QuickStego and Echo

Answer -

Steganography is the technique of hiding secret information inside files such as images, audio, video, or text so that others cannot detect the hidden message.

Types of Steganography

1. Image Steganography

Secret information is hidden inside image files.

2. Audio Steganography

Data is hidden inside audio files or sound signals.

3. Video Steganography

Secret data is hidden inside video files.

4. Text Steganography

Information is hidden inside text documents using special patterns or formatting.

QuickStego

QuickStego is a steganography tool used to hide text messages inside image files. It provides a simple way to secure hidden information.

Echo

Echo is an audio steganography tool used to hide secret data inside audio files without noticeable changes in sound quality.

---

5. Explain Keylogger Tool

Answer -

A keylogger is a type of monitoring software or hardware that records every key pressed on a keyboard. Attackers use keyloggers to steal usernames, passwords, banking details, and personal information.

Types of Keyloggers

1. Software Keylogger

Installed as a program that secretly records keyboard activity.

2. Hardware Keylogger

A physical device connected between the keyboard and computer to capture keystrokes.

Risks of Keyloggers

- Theft of passwords
- Financial fraud
- Privacy violations
- Data theft

Prevention Methods

- Install antivirus software
- Avoid suspicious downloads
- Use virtual keyboards
- Keep systems updated

Malware

1. Define Types of Viruses

Answer -

A computer virus is a malicious program designed to infect systems, damage files, steal information, or disrupt computer operations.

Types of Viruses

1. Boot Sector Virus

Infects the boot sector of storage devices and affects system startup.

2. File Infector Virus

Attaches itself to executable files and spreads when the file is opened.

3. Macro Virus

Infects documents and spreads through applications like Microsoft Word or Excel.

4. Multipartite Virus

Infects both boot sectors and files, making it difficult to remove.

5. Polymorphic Virus

Changes its code repeatedly to avoid detection by antivirus software.

6. Resident Virus

Stays in the computer memory and infects files automatically.

---

2. Explain Trojan Tool

Answer -

A Trojan Horse is a type of malware that appears to be legitimate software but secretly performs malicious activities in the background.

Trojans are commonly used to:

- Steal sensitive information
- Provide remote access to attackers
- Install additional malware
- Monitor user activity

Common Characteristics of Trojans

- Disguised as useful software
- Does not replicate like viruses
- Can create backdoors in systems

Prevention Methods

- Download software from trusted sources
- Use antivirus software
- Avoid opening suspicious files and emails

---

3. Explain Any One Antivirus with Example

Answer -

An antivirus is software designed to detect, prevent, and remove malware such as viruses, worms, ransomware, and spyware from computer systems.

Example – Windows Defender

Windows Defender is a built-in antivirus provided by Microsoft for Windows operating systems.

Features of Windows Defender

- Real-time protection
- Malware scanning

- Firewall integration
- Threat detection and removal
- Automatic security updates

Advantages

- Free with Windows
- Easy to use
- Provides continuous protection against threats